



*A Project Presentation
on*

Sentinel-X: AI-Based Network Attack Forecasting from Network Traffic Data

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Project Content Roadmap



01. Introduction & Shift

Reactive NIDS vs Predictive
Forecasting (SIH26153)

02. Literature & Objectives

Survey of existing NIDS & research
gaps addressed

03. System Pipeline

End-to-end telemetry workflow &
architecture

04. 16 State Features

10s aggregation capturing attack
kinetics

05. Sequence Windowing

Leak-free chronological 70/15/15
splitting

06. LSTM World Model

2-layer backbone with 3 multi-task
heads

07. Autoregressive Rollout

Multi-step rollout ($T+1..T+3$) & joint loss

08. MITRE ATT&CK Lifecycle

5-stage conservative behavioral
progression

09. Explainable AI (SHAP)

Signed risk drivers, inhibitors & temporal
saliency

10. Empirical Benchmarks

100% Recall, 75% FPR reduction &
degradation

11. SOC Command Dashboard

5 interactive views running locally offline

12. Relevance & Conclusion

Critical defense, green AI & future SDN
roadmap



Problem Statement: Shift to Predictive Defense



TRADITIONAL NIDS (REACTIVE)

- 1. Attacker launches exploit payload
- 2. Perimeter breached / system compromised
- 3. Alert fires POST-COMPROMISE (Too Late!)
- **Result: High MTTR & severe damage done**

SENTINEL-X (PREDICTIVE FORECASTING)

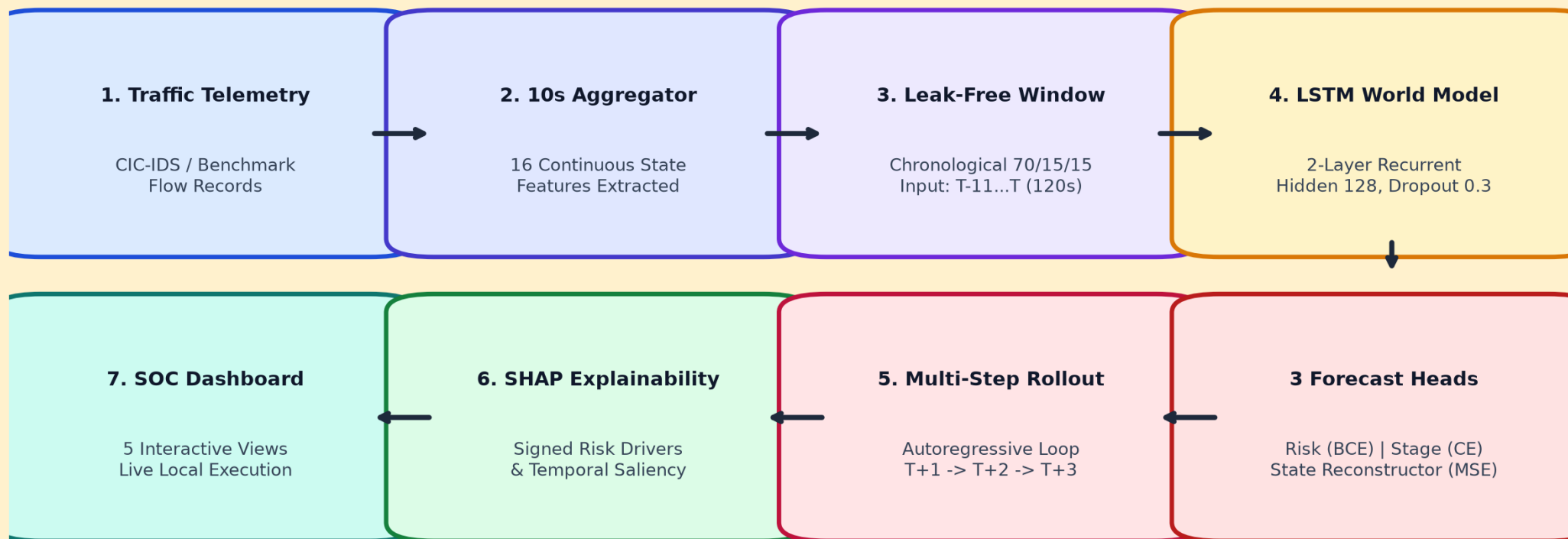
- 1. Monitors 120s continuous telemetry window
- 2. LSTM World Model forecasts 10-30s ahead
- 3. Alert & MITRE stage predicted PRE-BREACH
- **Result: 10-30s proactive mitigation buffer**

SIH26153 Mandate: Transition from reactive detection (alerting post-breach) to predictive forecasting.

- By predicting risk 10 to 30 seconds into the future, automated firewall policies can isolate malicious hosts before damage occurs.



End-to-End System Pipeline



Closed-Loop Data & Neural Pipeline: Raw Flows → 10s Window Aggregator → 2-Layer LSTM → 3 Heads → Rollout → Dashboard



16-Dimensional Network State Representation

VOLUMETRIC (5 Features)

- `flow_count`
- `total_packets`
- `packet_rate`
↳ Detects volumetric floods & exfiltration
- `total_bytes`

- `byte_rate`

PACKET DYNAMICS (2 Features)

- `fwd_pkt_len_mean`
- `bwd_pkt_len_mean`

↳ Detects payload size & auth asymmetry

TIMING & DURATION (3 Features)

- `duration_mean`
- `duration_std`
- `iat_mean` (inter-arrival)
↳ Detects rapid scanning & botnet pacing

FLAGS & PROTOCOLS (6 Features)

- `syn_ratio`, `rst_ratio`
- `ack_ratio`, `fin_ratio`
- `tcp_ratio`, `udp_ratio`
↳ Detects SYN scans, closed ports & floods

Why 10-Second Windows? Packet-by-packet sniffing overflows memory on gigabit links; statistical windowing captures attack kinetics.

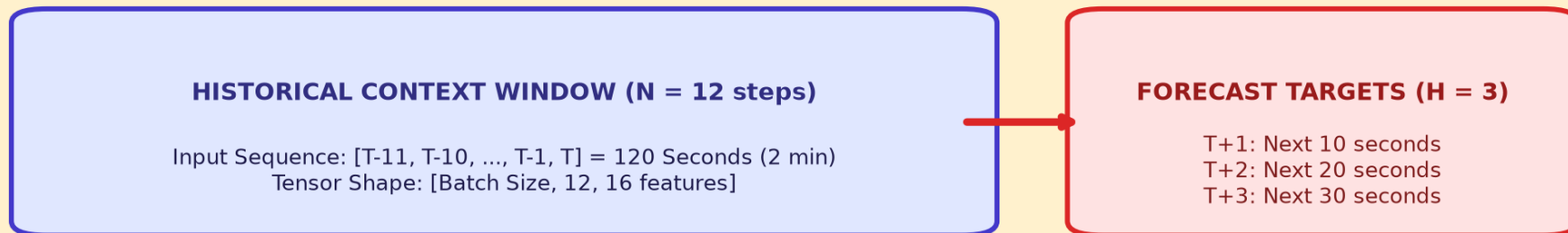


Leak-Free Chronological Windowing

LEAK-FREE CHRONOLOGICAL PARTITIONING (NO RANDOM SHUFFLING)



ROLLING SEQUENCE WINDOW STRUCTURE

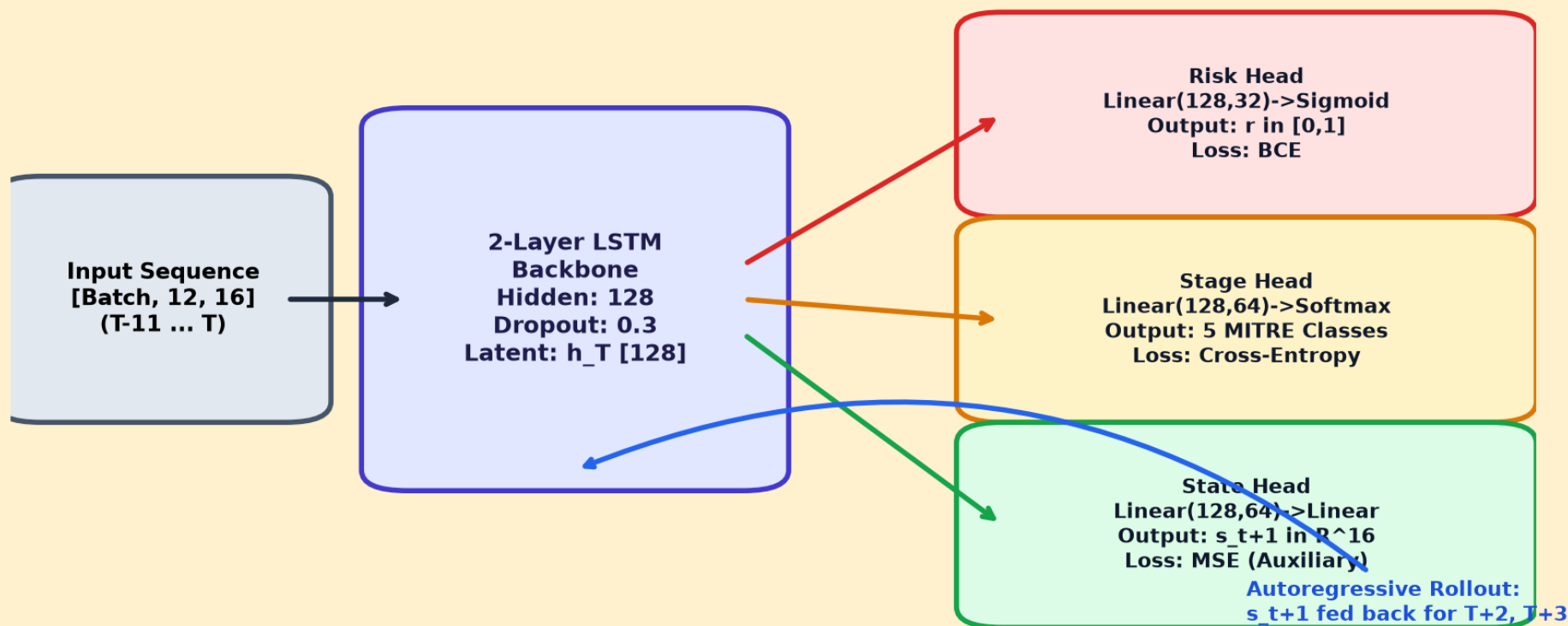


Guaranteed Zero Data Leakage: Standard `train_test_split(shuffle=True)` causes temporal leakage in time series.

- Strict chronological split (Train 70% | Val 15% | Test 15%). Scaler is fitted strictly on the training set.



Deep LSTM World Model Architecture



World Model Paradigm: It doesn't just output a static class; it predicts how the environment itself evolves.

- 2-Layer LSTM (Hidden 128) concurrently drives: Risk Head (BCE), Stage Head (CE), and State Reconstructor (MSE).



Autoregressive Rollout & Multi-Task Loss

AUTOREGRESSIVE MULTI-STEP ROLLOUT

STEP 1: T+1 (Next 10s)

Input: History [T-11 ... T]

Output: Predicted State s_{T+1} , Risk r_{T+1} , Stage c_{T+1}

STEP 2: T+2 (Next 20s)

Input: Predicted State s_{T+1} fed back

Output: Predicted State s_{T+2} , Risk r_{T+2} , Stage c_{T+2}

STEP 3: T+3 (Next 30s)

Input: Predicted State s_{T+2} fed back

Output: Predicted State s_{T+3} , Risk r_{T+3} , Stage c_{T+3}

WEIGHTED MULTI-TASK LOSS

$$L_{\text{total}} = 1.0 * L_{\text{BCE}} + 1.0 * L_{\text{CE}} + 0.2 * L_{\text{MSE}}$$

1. Risk Loss ($w = 1.0$)

Binary Cross-Entropy over risk probability [0, 1]

2. Stage Loss ($w = 1.0$)

Categorical Cross-Entropy over 5 MITRE stage classes

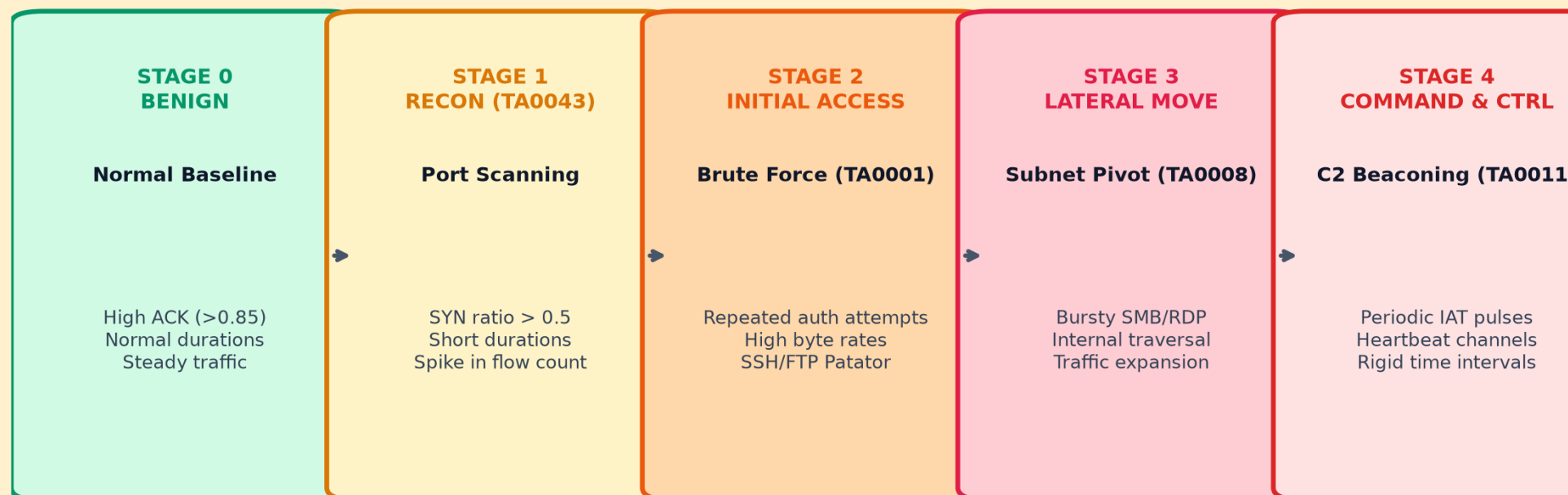
3. State Loss ($w = 0.2$)

Mean Squared Error on 16 continuous telemetry features
Acts as Auxiliary Regularization to avoid label overfitting

Auxiliary Task Regularization: Predicting continuous state ($w_{\text{state}}=0.2$) forces the LSTM to learn real network physics.



Conservative MITRE ATT&CK Lifecycle

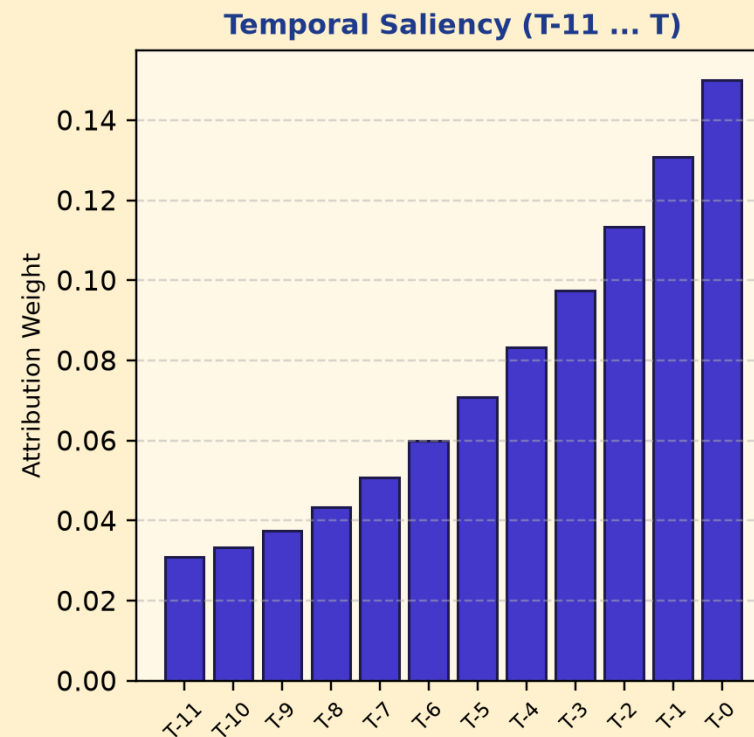
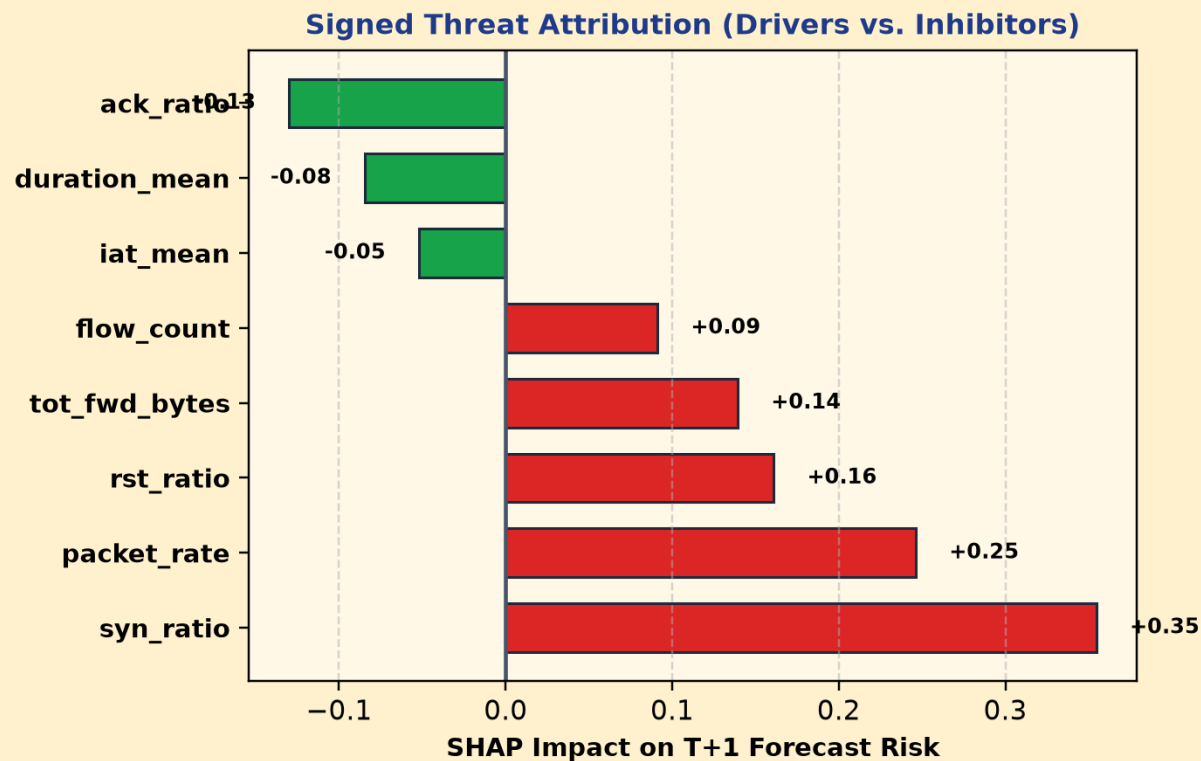


Academic Defensive Posture: We classify behavioral telemetry stages for SOC triage, NOT definitive forensic certainty.

- 5 Distinct Stages: Benign Baseline (0) → Recon (1) → Initial Access (2) → Lateral Movement (3) → Command & Control (4).



Explainable AI (SHAP Threat Attribution)



Eliminating the Black Box: Explains exactly WHY future risk was predicted to escalate.

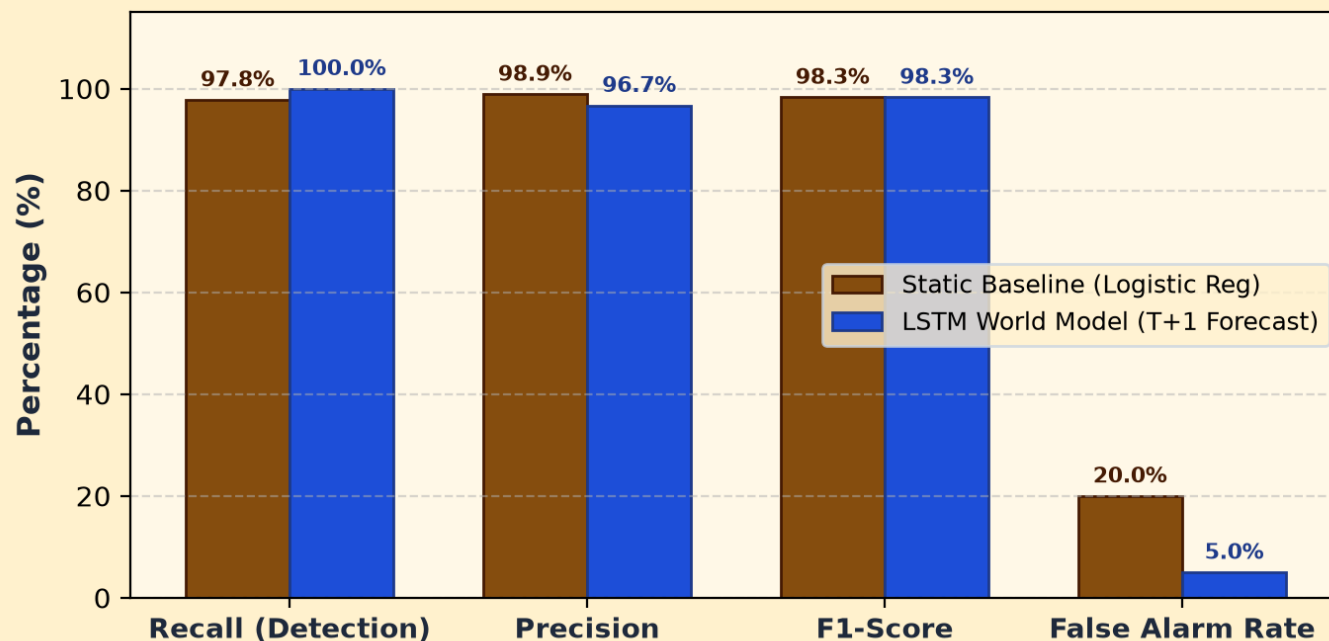
- Red Bars = Risk Drivers (syn_ratio, packet_rate) | Green Bars = Risk Inhibitors (ack_ratio, duration_mean).



Empirical Results: Baseline vs. World Model



Verified Benchmark: Baseline vs. Deep LSTM World Model



100.0% RECALL

Zero missed attacks at T+1
(Baseline missed 2.2%)

75% LESS FALSE ALARMS

FPR reduced from 20% to 5%
Directly solves alert fatigue

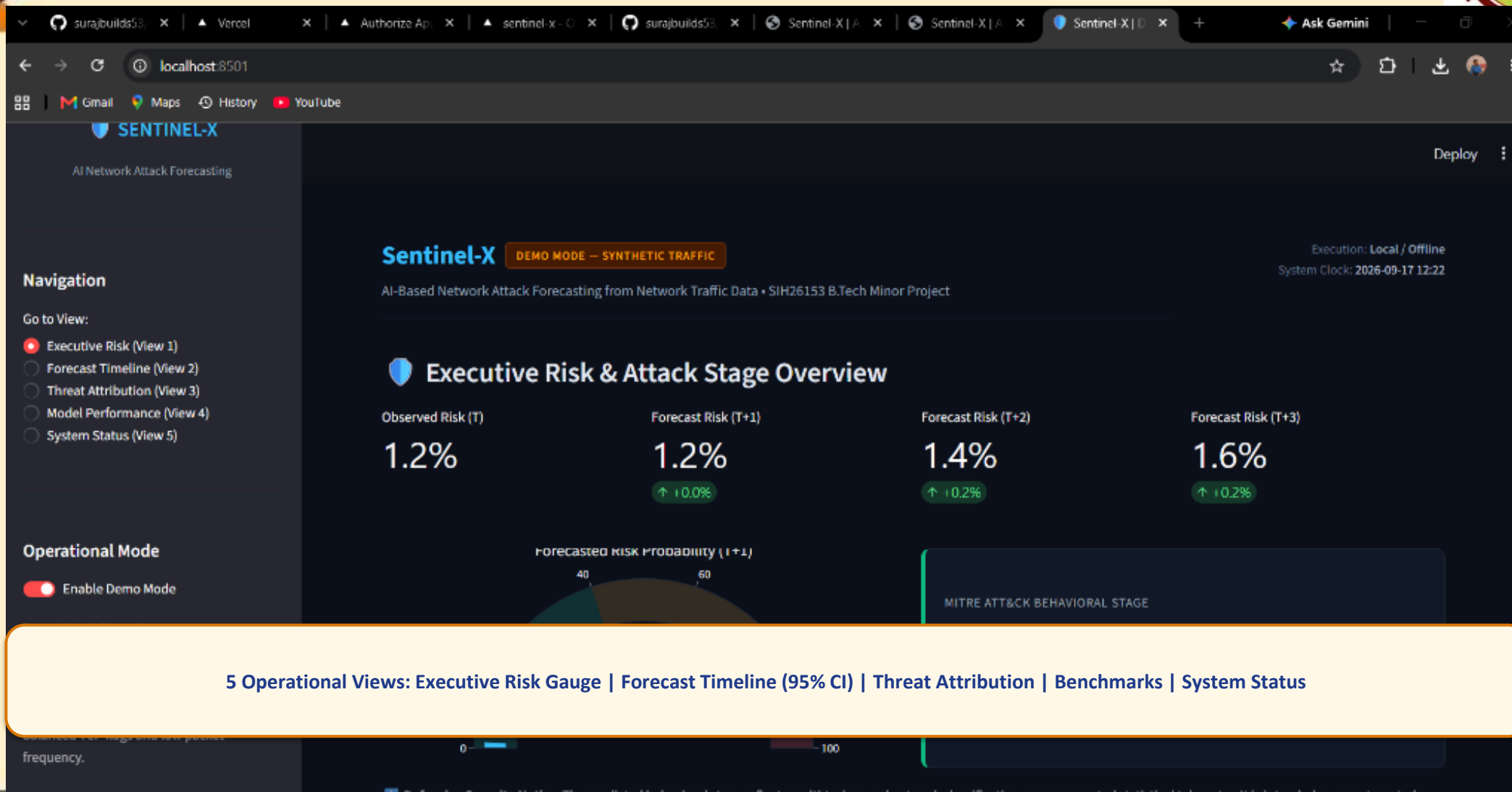
0.9834 F1-SCORE

F1 tapers to 0.9721 at T+3
(Expected horizon decay)

Key Takeaway: Predictive variance compounds naturally as horizon extends, proving genuine autoregressive rollout.



Interactive SOC Command Dashboard





Societal Impact & Future Roadmap



AUTOMATED SDN INTEGRATION

Direct API linkage to OpenFlow / SDN controllers to push automatic firewall ACLs upon high-confidence 1+1 risk alarms.

TEMPORAL GRAPH NETWORKS

Integrating Graph Neural Networks (GNNs) to model host-to-host lateral traversal topology alongside sequential time-series.

EDGE ROUTER QUANTIZATION

Quantizing the PyTorch LSTM model using ONNX / TensorRT for sub-millisecond deployment inside edge enterprise routers.

Green AI Architecture: Single-sample inference latency < 20 ms on standard laptop CPU with zero cloud costs.

- Proactive shielding for critical infrastructure (power, banking, healthcare) against multi-phase ransomware & breaches.



THANK YOU

Questions & Discussion

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GitHub: github.com/surajbuilds53/sentinel-x | Live Hub: sentinel-x-lilac.vercel.app